Code structure

js/util/solvePlanarPose.js — Recovers the 3D pose of a rectangle from its 4 corners in an image.

js/util/transformMatrix.js — Helper for 4×4 matrix operations.

js/util/screenAnchor.js — The util scenes import. Handles the calibration popup, sliders, capture, polling, sync, and lock.

js/scenes/transferCalibrated.js — First scene using the util — a copy of transfer.js with the pane anchored to the real screen.

python/aruco_detector.py — Detection logic. Given an image, finds the markers. Wraps cv2.aruco to detect markers in a frame.

python/screen_anchor_server.py — HTTP server. Receives frames from the PC, runs them through aruco_detector.py, and persists calibration to JSON.

screenAnchorConfig.json — Auto-generated at project root. Stores last-used calibration so you don't re-tune each session.

ArUco markers

They come from OpenCV. Think of them as simpler, more precise QR codes — widely used in robotics for fast and reliable pose tracking.

How to use it

Setup (once)

1. Install dependencies: pip install opencv-contrib-python numpy.
   It has to be opencv-contrib-python, not opencv-python. The ArUco module only ships in the contrib version.

Every session

2. Start the Python server: cd python && python screen_anchor_server.py

3. Open the scene on the PC. The registration panel appears top-right. A white popup with four black markers opens, that's the registration window.

4. Set WIDTH and HEIGHT to the real physical size of that popup window. Measure it with a ruler if you need to. This is what the solver uses to reconstruct depth.

5. Click Start Capture. Select the window showing the headset's cast (Meta Quest Developer Hub or browser cast). The PC starts sending frames to the server every 500ms.
   * If Meta Quest Developer Hub cast is used, select “left eye” in the cinematic view.

6. Put on the headset and aim it at the screen until all four markers are visible. When the server detects them all, the Lock Calibration button turns green.

7. Hold still and click Lock Calibration on the PC. The headset solves the pose using its head position at that exact moment.

Done. The registration popup closes, the 3D pane is now anchored to the physical screen. Settings are saved automatically, next session the sliders start from your last values.

If something looks off

- Pane at wrong depth → adjust Focal Length and re-lock.
- Screen moved or results drifted → press R (or Recalibrate) and redo from step 6.